MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

SCHOOL OF COMPUTER ENGINEERING

V SEMESTER B.TECH. (CCE)

ICT 3128 Wireless Communication and Computing

Date: 29-08-2026 FISAC-2: Group Assignment Marks: 05

Assignment Submission Instructions

1. **Assignment Details:** Mention the following details on the assignment sheet:
   - Subject Code
   - Subject Name
   - Date
   - Section
   - Group Number
   - Registration Number of each group member
   - Names of all group members
2. **Contribution Details:** The assignment sheet must include a **table showing the contribution of each group member**.
3. **Handwritten Assignment:** The assignment must be **handwritten**. Each student must write their **name and signature on every page**. The completed assignment should be scanned, and the required pictures/screenshots should be inserted at the appropriate locations.
4. **Individual Contribution:** Each team member must complete **one sub-question entirely**. For example:
   - Person 1 – Question 1A
   - Person 2 – Question 1B
   - Person 3 – Question 1C
   - Person 4 – Question 1D

   The **Team Lead** must check the overall flow and correctness of the report and consolidate the contributions of all team members into **one PDF file**.
5. **Submission Format**:
   - Only the Team Lead should upload the assignment.
   - Upload only one PDF file.
   - PDF file name: GroupNumber_QuestionNumber
   - Example: G01_Q2.pdf
6. **Last Date for Submission: 25/10/2026 (25 October 2026), on or before 5:00 PM**

Important: Please ensure that the assignment is prepared and submitted strictly according to the above instructions. Submissions that do not follow the specified format may not be considered for evaluation.

MATLAB Software: Meet Lab instruction Mrs. Tulasi, Computing Lab -15, 1st Floor, SCE, AB-5, MIT

Common Project Requirements

- Every question has the same workload: 4 transmitter tasks, 2 channel/propagation tasks, 2 receiver tasks and 3 analysis/engineering-decision tasks.
- All projects must implement or explicitly model a complete transmitter - channel/environment - receiver chain.
- Use a reproducible payload; where registration-number-derived data is requested, document the conversion and framing method.
- Include quantitative validation using appropriate metrics such as BER, PER, EVM, SINR, outage, throughput, latency, capacity, spectral efficiency or link margin.
- Where Mobile apps/sniffers are used, analyze only permitted measurements and the group's own traffic and follow institutional/network policies.

Mobile APP		**Mobile APP**
Network signal guru		CellMapper
G-NetWiFi		NetMonitor cell Signal Logging
Network Cell Info		DW-RF Mobile Trace
Network Signal Info lite		WiFi Analyzer
NetMonster		WiFi Monitor network analyzer
LTE Discovery (5G NR)		Cell Signal Monitor
Network survey		RF Signal Tracker and Detector
4G LTE/5G Coverage monitor		Signal Detector (lefan Co. Ltd)
Cellular-Z		G-Net Tracker Lite
Analiti		Tower Collector

- Each project must end with a justified engineering decision, not only plots or screenshots.

20. Integrated Wireless Capstone: Adaptive Cellular/WLAN Transceiver with Coverage, Mobility and Capacity Validation

BTL6 - Create

Syllabus: Wireless system design; Link budget; Fading; Cellular concepts; OFDM/MIMO; WLAN; Coverage and capacity

Project scenario: Create an integrated end-to-end wireless prototype that combines a realistic transmitter, channel/environment and receiver and uses at least one field-measurement input from Mobile Apps (page-2). The project shall demonstrate how physical-layer design, coverage and user performance interact in one complete system.

A. Transmitter design and implementation

- Generate framed traffic with preamble/payload/CRC and implement an OFDM-based transmitter using adaptive QPSK/16-QAM/64-QAM or a justified equivalent.
- Include configurable transmit power, antenna gain, channel bandwidth and optional 2x2 MIMO mode.
- Use one permitted field-measurement dataset to calibrate or validate a link parameter such as path loss, RSSI trend, channel occupancy or latency.
- Plot transmitter spectrum/resource mapping and calculate nominal throughput and EIRP/link parameters.

B. Channel / propagation model

- Model path loss, shadowing, multipath fading, AWGN and one additional impairment such as interference or mobility/Doppler.
- Evaluate a scenario containing at least three receiver locations or mobility states and two traffic/load conditions.

C. Receiver design and implementation

- Implement synchronization, channel estimation/equalization, adaptive demodulation and CRC/PER measurement; include MIMO detection if selected.
- Recover the payload and log received power/SINR, BER/PER, throughput and latency/outage proxy across the scenario.

D. Analysis, validation and engineering decision

- Validate at least one simulated trend against the mobile apps (page-2) measurement and explain deviations.
- Identify the dominant limitation among coverage, interference, mobility, bandwidth and receiver performance.
- Produce a final engineering configuration and justify it against explicit reliability, throughput and coverage targets.

Required project evidence: Complete MATLAB/Simulink project, end-to-end architecture, field evidence, parameter/link-budget tables, Tx/Rx plots, BER/PER/throughput/coverage results and final engineering design justification.